

Simulation Applicability Context Module (SACM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

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Governed Space: Simulation Applicability Context

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Governed Space

Simulation Applicability Context

Canonical Definition

Simulation Applicability Context Module (SACM) defines the governance conditions under which the operational, environmental, temporal, spatial, population, configuration, capability, behavioral, dependency, scenario, and decision contexts to which a simulation inquiry is intended to apply are explicitly identified, bounded, documented, and maintained throughout the simulation lifecycle.

SACM establishes the contextual envelope within which the simulation is intended to produce meaningful simulation outputs.

It prevents a simulation designed for one context from being interpreted as if it represented every context in which the Simulation Object may exist.

Operational Role

SACM serves as the fourth internal governance space of the Simulation Purpose & Scope Standard (SPSS).

The preceding modules establish:

SPM — Why is the simulation being performed?

SSCM — What exactly must the simulation cover?

SEXM — What remains outside the governed Simulation Scope?

SACM asks:

Within which contexts is the governed simulation inquiry intended to apply?

The governed progression is:

Simulation Purpose → Simulation Scope → Simulation Exclusions → Applicability
Dimensions → Applicability Conditions → Applicability Boundary → Applicability
Context

SACM therefore transforms a bounded Simulation Scope into an explicitly
contextualized simulation inquiry.

Module Operational Space

SACM governs: Simulation Applicability Context, applicability dimensions, applicability conditions, applicability boundaries, operational applicability, environmental applicability, spatial applicability, temporal applicability, population applicability, configuration applicability, capability applicability, behavioral applicability, interaction applicability, autonomy applicability, scenario applicability, dependency applicability, decision-context applicability, consequence-context applicability, organizational applicability, infrastructure applicability, applicability assumptions, applicability dependencies, applicability combinations, applicability restrictions, applicability uncertainty, conditional applicability, partial applicability, applicability conflicts, applicability change, applicability expansion, applicability narrowing, applicability traceability, and Applicability Context status.

Module Function

A simulation may be properly scoped and correctly executed while still being misused because its results are applied outside the context for which the simulation was designed.

SACM exists to prevent:

Simulation Performed in Context A → Results Produced → Context Forgotten →
Results Applied to Context B

without a governed basis.

Its function is to preserve the relationship:

Simulation Purpose + Scope + Exclusions → Intended Context → Simulation Design
→ Simulation Outputs → Context-Bounded Interpretation

SACM does not determine whether the simulation is ultimately valid for
operational reliance.

It establishes the context the simulation is designed to address.

Simulation Applicability Context

A Simulation Applicability Context is the defined combination of conditions within which the simulation inquiry is intended to represent, examine, or explore the Simulation Object.

The context may include:

Where

When

Under Which Conditions

For Which Population

For Which Configuration

At Which Capability State

Under Which Behaviors

Within Which Dependencies

For Which Scenario Class

and:

For Which Intended Decision Context Applicability Context Is Multidimensional

Simulation applicability should not normally be represented as a single universal label.

A simulation may simultaneously be:

Applicable — Environment A

Applicable — Configuration 3.2

Applicable — Population X

Applicable — Normal Operation

Not Intended — Emergency Operation

Not Intended — Configuration 4.0

The resulting applicability envelope is multidimensional. Applicability Context $\neq$ Simulation Scope

Simulation Scope and Simulation Applicability Context are related but distinct.

Simulation Scope asks:

What must the simulation examine?

Simulation Applicability Context asks:

Under which contextual conditions is that inquiry intended to apply?

For example:

Scope: Autonomous emergency braking performance.

Applicability Context: Passenger vehicles operating on paved European roads at defined speeds and environmental conditions.

The first defines the inquiry.

The second defines its intended contextual envelope. Applicability Context ≠ Simulation Exclusion

An exclusion identifies what remains outside the governed Simulation Scope.

An applicability boundary identifies where the simulation inquiry is intended to apply.

These may interact without becoming identical.

For example:

Snow explicitly excluded

may create:

Applicability restricted to non-snow conditions.

SEXM governs the exclusion.

SACM governs the resulting applicability context. Applicability Context ≠ Operational Reality

ORA™ governs actual Operational Reality.

SACM does not determine what conditions currently exist in the real world.

It governs:

the contextual conditions for which the simulation is designed.

The relationship may later be examined as:

Simulation Applicability Context ↔ Actual Operational Reality

but the two spaces remain architecturally distinct.

Operational Applicability

Operational applicability identifies the operating conditions addressed by the simulation.

Examples may include: normal operation, degraded operation, emergency operation, maintenance state, startup, shutdown, autonomous operation, supervised operation, remote operation, or defined mission phases.

A simulation intended for nominal operation should not silently acquire emergency-operation applicability.

Environmental Applicability

Environmental applicability establishes which environmental conditions form part of the intended simulation context.

Examples may include: temperature range, weather, visibility, radiation, terrain, atmospheric conditions, electromagnetic environment, network conditions, traffic density, industrial environment, or other environmental factors.

Spatial Applicability

Spatial applicability defines the physical, geographic, virtual, or logical spaces for which the simulation inquiry is intended.

Examples include:

specific facility

defined road network

airspace class

orbital region

geographic region

digital network environment

or:

virtual operational domain

Temporal applicability identifies time-related contextual boundaries.

It may govern: simulation period, operational duration, mission phase, lifecycle stage, seasonal conditions, time horizon, system age, or another temporally bounded condition.

A simulation of short-duration system behavior must not automatically imply long-duration applicability.

Population Applicability

Where humans, users, organizations, agents, vehicles, devices, or other populations are relevant, SACM identifies which population characteristics belong to the intended context.

Examples:

Adult Population

Pediatric Population

Trained Operators

General Public

Vehicle Class A

Robot Type B

Population applicability prevents results generated for one defined population from being automatically generalized to another.

Configuration Applicability

A simulation may apply only to a particular: hardware configuration, software version, model version, component combination, system architecture, parameter set, or deployment configuration.

SACM preserves this contextual binding.

Capability Applicability

Simulation applicability may depend upon the capability state of the Simulation Object.

For example:

AI advisory capability

may be represented while:

autonomous tool execution

is outside the intended applicability context.

A material capability change may therefore invalidate or narrow the applicability of the existing simulation inquiry.

Behavioral Applicability

A simulation may address only defined behavioral modes.

Examples include:

Nominal Behavior

Cooperative Behavior

Adversarial Behavior

Human-Supervised Behavior

Autonomous Behavior

Failure Behavior

Behavioral applicability should remain explicit wherever behavioral differences materially affect the simulation inquiry.

Interaction Applicability

Where multiple actors or systems interact, applicability may depend upon defined interaction structures.

Examples: human–AI interaction, robot–human interaction, vehicle–vehicle interaction, multi-agent coordination, infrastructure–system interaction, or cross-system communication.

Representation of the actors alone does not establish applicability to every possible interaction among them.

Autonomy Applicability

For autonomous systems, the degree of autonomy may materially define applicability.

A simulation performed under:

Human-in-the-Loop

conditions cannot automatically support applicability to:

Fully Autonomous Operation.

SACM preserves the applicable autonomy context.

Scenario Applicability

A simulation may apply to a defined scenario class.

Examples include:

Nominal Scenarios

Edge Cases

Emergency Scenarios

Adversarial Scenarios

Rare Events

High-Density Operations

Failure & Recovery Scenarios

The applicability record should distinguish the scenario classes actually intended by the simulation inquiry.

Dependency Applicability

Simulation meaning may depend upon external conditions or systems.

Examples include: network availability, sensor infrastructure, positioning services, human supervision, external data, power availability, third-party services, or supporting infrastructure.

Applicability may therefore be conditional upon those dependencies remaining present.

Infrastructure Applicability

Where simulation outcomes depend upon infrastructure characteristics, the applicable infrastructure context should be identified.

For example:

Warehouse Infrastructure Type A

may differ materially from:

Open Industrial Yard

even when the same autonomous robot is simulated.

Decision-Context Applicability

A simulation may be designed to inform a specific class of decisions.

Examples include: design selection, engineering analysis, safety investigation, mission planning, operational planning, capacity planning, training, policy exploration, validation evidence generation, or research.

A simulation designed for exploratory research should not automatically be treated as suitable for high-consequence operational authorization.

SACM identifies the intended decision context.

It does not itself authorize reliance.

Consequence Context

The consequence of using simulation outputs may affect how narrowly applicability should be established.

A simulation informing:

early concept exploration

may operate under a different applicability requirement than one intended to support:

safety-critical deployment decisions.

SACM preserves the intended consequence context so downstream governance can apply the appropriate requirements.

Applicability Conditions

A Simulation Applicability Condition is a condition that must hold for the intended contextual relationship between the simulation and its governed inquiry to remain applicable.

Examples:

Software Version = 4.2

Temperature = -10°C to $+40^{\circ}\text{C}$

Human Supervisor Present

Maximum Payload ≤ 500 kg

Network Latency $\leq$ Defined Threshold

Applicability conditions should be explicit where their violation materially changes simulation meaning.

Conditional Applicability

A simulation may be applicable only while specific conditions remain true.

This may be represented as:

Conditionally Applicable

The condition should remain traceable to the relevant applicability dimension.

Partial Applicability

A simulation may apply to only part of the intended operational context.

For example:

Applicable — Normal Operation

Partially Applicable — Degraded Operation

Not Intended — Emergency Operation

Partial applicability should remain visible rather than being collapsed into a universal applicability statement.

Applicability Boundary

The Simulation Applicability Boundary separates:

Context Addressed by the Simulation

Inquiry

from:

Context Not Addressed by the Simulation

Inquiry

This boundary may be: operational, environmental, spatial, temporal, population-based, configuration-based, capability-based, behavioral, scenario-based, dependency-based, or multidimensional.

Applicability Envelope

Multiple applicability dimensions may be combined into a Simulation Applicability Envelope.

Conceptually:

Environment × Time × Population × Configuration × Capability × Behavior × Scenario × Dependencies

defines the contextual space in which the simulation inquiry is intended to operate.

The envelope is conceptual and need not be numerical.

Applicability Combination

Individual applicability conditions may interact.

For example:

Rain

may be individually included.

Night

may be individually included.

But:

Night + Heavy Rain

may not have been included as a combined simulation context.

SACM therefore distinguishes:

Individual Context Applicability

from:

Combined Context Applicability

where interaction effects are material.

No Applicability by Component Rule

The fact that individual conditions were represented separately does not automatically establish applicability to every combination of those conditions.

Applicability Assumption

An applicability context may depend upon assumptions.

For example:

Communication infrastructure assumed continuously available.

The applicability record should preserve the dependency on that assumption.

The assumption itself is governed by the appropriate downstream SIMULOS® assumption governance.

Applicability Dependency

An applicability context may depend upon another condition remaining valid.

Conceptually:

Applicability Context → Required Dependency → Dependency State

Where the dependency materially changes, applicability may require reassessment.

Applicability Restriction

A Simulation Applicability Restriction explicitly prevents interpretation outside a defined contextual boundary.

Examples:

Not applicable to outdoor operation

Not applicable above defined velocity

Not applicable to pediatric population

Not applicable to fully autonomous operation

Restrictions should remain visible in downstream simulation interpretation.

Applicability Uncertainty

Sometimes it may be unclear whether a simulation remains applicable to a context.

Such uncertainty should not be converted automatically into positive applicability.

Possible status:

Applicability Undetermined

until sufficient governance information exists.

Applicability Conflict

An applicability conflict exists where different parts of the governed simulation record imply incompatible applicability boundaries.

For example:

Simulation Scope: Global operation.

Environmental Applicability: Temperate climates only.

The conflict should be resolved or explicitly preserved before broad simulation claims are made. Applicability–Purpose Alignment

The applicability context must remain capable of supporting the Simulation Purpose.

If the purpose requires:

global autonomous navigation safety

but applicability is limited to:

daytime operation on one mapped campus,

the mismatch is material.

SACM should expose that mismatch rather than allowing the narrow simulation context to satisfy the broader purpose implicitly. Applicability–Scope Alignment

Applicability must remain consistent with the governed Simulation Scope.

A scope describing multiple environments cannot be treated as satisfied where the applicability context covers only one environment unless the difference is explicitly governed. Applicability–Exclusion Alignment

Material exclusions should be reflected in applicability where appropriate.

Conceptually:

Material Exclusion → Applicability Consequence

For example:

Emergency Operation Excluded

should prevent the applicability context from silently including emergency operation.

No Universal Applicability Rule

No simulation should be treated as universally applicable merely because its applicability boundary was not documented.

Undefined applicability is not universal applicability.

No Context Transfer Rule

A simulation designed for Context A must not automatically acquire applicability to Context B merely because the Simulation Object is the same.

No Context Generalization Rule

Simulation outputs must not be generalized beyond the governed applicability context without an additional basis.

No Configuration Transfer Rule

Results associated with one materially relevant configuration must not automatically transfer to another materially different configuration.

No Population Transfer Rule

Results associated with one population must not automatically be generalized to another materially different population.

No Environment Transfer Rule

Results associated with one environment must not automatically be treated as applicable to another materially different environment.

No Autonomy Transfer Rule

Simulation under one autonomy condition must not automatically establish applicability under a materially different autonomy condition.

No Scenario Inflation Rule

Simulation of selected scenarios must not be represented as applicability to all possible scenarios.

Applicability Change

Applicability may change when the simulation inquiry, Simulation Object, assumptions, model, environment, scenario structure, or operational context changes.

SACM therefore treats applicability as a governed contextual state rather than permanent metadata.

Applicability Expansion

An applicability context may expand when new simulation work sufficiently addresses additional: environments, populations, configurations, scenarios, capabilities, behaviors, or dependencies.

Expansion should have a traceable basis.

Applicability Narrowing

New information may demonstrate that the simulation applies to less than originally expected.

Narrowing is a legitimate governance outcome.

For example:

Originally Intended: Urban Roads

may become:

Supported Simulation Context: Urban Roads Below 60 km/h

after limitations are identified.

SACM permits the applicability context to become narrower without treating this as methodological failure.

Applicability Change Trigger

Material changes that may require applicability reassessment include:
Simulation Object version change, configuration change, new capability, new autonomy level, new environment, new population, new scenario class, changed

dependency, changed purpose, changed scope, or changed exclusions.

Applicability Status

Possible statuses may include:

Applicable Context Established

Conditionally Applicable

Partially Applicable

Applicability Restricted

Applicability Under Review

Applicability Conflicted

or:

Applicability Undetermined Applicability Context Record

A governed simulation may maintain a Simulation Applicability Context Record identifying: Simulation Object, Simulation Purpose, Simulation Scope, material exclusions, applicable operational conditions, applicable environments, spatial boundaries, temporal boundaries, applicable populations, configurations, capability states, behavioral modes, autonomy conditions, scenario classes, dependencies, decision context, consequence context, applicability conditions, restrictions, uncertainties, status, and change history.

Applicability Traceability

Material applicability conditions should remain traceable through the simulation lifecycle.

A downstream reviewer should be capable of determining:

For which context was this simulation actually designed?

and:

Does the current interpretation remain inside that context?

Failure Conditions

SACM governance failure may exist where: applicability context is undefined, a bounded simulation is represented as universally applicable, applicability conflicts with Simulation Purpose, applicability conflicts with Simulation Scope, material exclusions are not reflected in applicability, results are transferred between materially different environments without basis, population differences are ignored, configuration changes are ignored, autonomy changes are ignored, scenario applicability is inflated, dependencies

affecting applicability remain undocumented, combined-context effects are ignored, applicability uncertainty is represented as established applicability, applicability expansion occurs without supporting simulation work, applicability restrictions disappear from downstream interpretation, or simulation outputs are represented outside the context for which the simulation was designed.

Minimum Implementation Framework

1. Identify Applicability Dimensions

Determine the operational, environmental, spatial, temporal, population, configuration, capability, behavioral, autonomy, scenario, dependency, and decision dimensions material to the simulation inquiry. 2. Establish Applicability Conditions

Define the contextual conditions under which the simulation is intended to apply. 3. Define the Applicability Boundary

Distinguish applicable, conditionally applicable, partially applicable, restricted, and undetermined contexts. 4. Align Applicability With Purpose, Scope, and Exclusions

Verify that the applicability context does not exceed the governed Simulation Purpose, Scope, or material exclusions. 5. Preserve Applicability Traceability

Maintain the Simulation Applicability Context Record and propagate material contextual boundaries into downstream simulation design, outputs, interpretation, and lifecycle reassessment.

Governance Outputs

SACM may produce: Simulation Applicability Context, Simulation Applicability Context Record, Applicability Dimension Map, Operational Applicability Record, Environmental Applicability Record, Spatial Applicability Record, Temporal Applicability Record, Population Applicability Record, Configuration Applicability Record, Capability Applicability Record, Behavioral Applicability Record, Interaction Applicability Record, Autonomy Applicability Record, Scenario Applicability Record, Dependency Applicability Record, Infrastructure Applicability Record, Decision-Context Applicability Record, Consequence-Context Applicability Record, Applicability Condition Record, Conditional Applicability Record, Partial Applicability Record, Applicability Restriction Record, Applicability Uncertainty Record, Applicability Dependency Map, Combined Context Applicability Record, Simulation Applicability Envelope, Applicability–Purpose Alignment Record, Applicability–Scope Alignment Record, Applicability–Exclusion Alignment Record, Applicability Conflict Record, Applicability Change Record, Applicability Expansion Record, Applicability Narrowing Record, Simulation Applicability Status, and Simulation Applicability Traceability Map.

Use Case 1 — Autonomous Vehicle Simulation

Scenario

An autonomous-driving simulation evaluates collision avoidance.

The Simulation Scope covers urban driving.

The simulation environment represents:

daylight,

dry and wet roads,

speeds up to 60 km/h,

and:

defined European road configurations.

Snow, severe fog, and speeds above 60 km/h are outside the governed inquiry.

Application

SACM establishes the Simulation Applicability Context as:

Urban European Road Environment

Daylight Operation

Dry / Wet Surface Conditions

≤ 60 km/h

The simulation is not assigned applicability to severe fog, snow, higher speeds, or materially different road environments. Result

The simulation outputs retain a clear contextual boundary.

The fact that collision avoidance performed successfully within the governed context cannot be used to imply equivalent simulation coverage across all driving environments. Use Case 2 — Spacecraft Autonomous Navigation Simulation

Scenario

A spacecraft navigation system is simulated for autonomous orbital insertion.

The simulation represents:

defined spacecraft configuration,

specified orbital conditions,

nominal communication availability,

and:

a bounded propulsion-performance range.

A later spacecraft configuration introduces a modified propulsion system.

Application

SACM binds the original Simulation Applicability Context to the original configuration and propulsion conditions.

The new propulsion configuration creates an Applicability Change Trigger.

The previous simulation is not automatically considered applicable to the modified spacecraft. Result

The simulation lifecycle preserves the distinction between:

same mission purpose

and:

different applicability context.

Additional simulation work can determine whether applicability may legitimately expand to the new configuration.

Architectural Position

Simulation Applicability Context Module (SACM) is the fourth internal governance space of the Simulation Purpose & Scope Standard (SPSS).

The internal progression is:

Simulation Purpose → Simulation Scope → Simulation Exclusions → Simulation Applicability Context → Simulation Depth & Precision

SPM establishes why the simulation exists.

SSCM establishes what it must cover.

SEXM establishes what remains outside the governed inquiry.

SACM establishes the contextual envelope within which that inquiry is intended to apply.

SDPM subsequently establishes how deeply and precisely the simulation must examine that governed inquiry.

SACM does not govern actual Operational Reality under ORA™, general validation applicability under VALIDOS®, integrity under INTEGROS®, or the quality and fidelity with which reality is represented under the subsequent Reality Representation Standard (RRS).

It governs specifically:

the contextual applicability boundary of the simulation inquiry.

Foundational Principle

A simulation can legitimately represent only the contextual space for which its inquiry was designed; applicability must therefore remain explicitly bounded wherever changes in environment, time, population, configuration, capability, behavior, scenario, or dependency could materially change the meaning of its results.

Canonical Closing Statement

Simulation Applicability Context Module (SACM) establishes the contextual boundary of governed simulation by defining the operational, environmental, spatial, temporal, population, configuration, capability, behavioral, autonomy, scenario, dependency, and decision conditions within which a simulation inquiry is intended to apply. Through applicability dimensions, contextual conditions, applicability envelopes, restrictions, dependency mapping, alignment with purpose, scope, and exclusions, anti-generalization rules, change governance, and lifecycle traceability, SACM prevents simulation outputs produced within one bounded context from silently acquiring meaning across materially different contexts. By preserving where, when, for whom, under which configuration, and under which conditions a simulation is intended to apply, SACM ensures that contextual boundaries remain part of the simulation itself rather than becoming invisible after results are produced.

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